Quantum State Discrimination with Bosonic Channels and Gaussian States

Course: PH3203 Advanced Quantum Mechanics
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A Term Paper Review of Chapters 1–3[1]

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Abstract

[ABSTRACT]

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1 Introduction

1.1 The notion of a quantum state

A quantum state represented by a positive Hermitian operator $\hat{\rho}$ with unit trace, contains all physically relevant information about a quantum system. If two quantum systems are prepared in identical states, any measurement performed on them will yield identical outcome probabilities, making the different preparation methods observationally equivalent. Conversely, if the systems are in different quantum states, there necessarily exists at least one measurement that produces different probability distributions, allowing their preparations to be distinguished. The systematic process of designing and performing measurements to identify which of several possible states (and thus which preparation) a given system is in is called quantum state discrimination.

1.2 Measuring a quantum state

While the quantum state itself is an abstract mathematical object, measurements provide it with concrete operational meaning. The outcomes of quantum measurements are always classical, such as a photodetector click count or an ion's position in a trap, and these outcomes behave as random variables. By performing the same measurement many times on identically prepared quantum systems, one can reconstruct the statistical distribution of outcomes. If this process is repeated for every possible measurement, the complete set of outcome probability distributions fully characterizes the quantum state. Thus, the state is not directly observable, but its empirical reality is revealed through the statistics of measurement outcomes, bridging the gap between abstract formalism and experimental practice. The formal relationship between quantum states and measurements is captured by the postulates of standard quantum measurement theory. These postulates can be summarized as follows:

1. Every physical observable corresponds to a Hermitian operator $\hat{X}$ with a spectral decomposition
$$\hat{X} = \sum_j \lambda_j |j\rangle\langle j| \tag{1}$$
where λ_j are the possible measurement outcomes.

2. The projectors
$$P_j = |j\rangle\langle j| \tag{2}$$
associated with the eigenstates form a complete set, satisfying
$$\sum_j P_j = \mathbb{I} \tag{3}$$

and are orthogonal such that

$$P_i P_j = \delta_{ij} P_i \tag{4}$$

3. A measurement of $\hat{X}$ yields one of its eigenvalues λ_j as the result.

4. If the outcome λ_j is obtained, the system collapses to the post-measurement state

$$\hat{\rho}_j = \frac{P_j \hat{\rho} P_j}{\mathrm{tr}(P_j \hat{\rho} P_j)} \tag{5}$$

5. The probability of obtaining the outcome λ_j is given by

$$p_j = \mathrm{tr}(\hat{\rho} P_j) \tag{6}$$

6. When the measurement is performed but the outcome is not recorded, the post-measurement state is described by the density operator

$$\hat{\rho}' = \sum_j P_j \hat{\rho} P_j \tag{7}$$

representing an average over all possible outcomes.

It is important to note that in stating the postulates, the Hilbert space has been assumed to be finite and discrete solely for the sake of notational simplicity. In principle, these postulates apply equally to states with infinite-dimensional Hilbert spaces, which is precisely the case for continuous variable systems. The postulates make it clear that the measurement process is fundamentally random, with the outcome able to take a range of values. The number of possible measurement outcomes is equal to the dimension of the Hilbert space of the state. The projection operators P_j used in this formulation are referred to as projective measurements or von Neumann measurements. However, it turns out that the requirement that measurement operators be strictly projective can be relaxed while still maintaining a meaningful statistical distribution for the outcomes. In particular, Postulate 3 provides an algorithm for generating outcome probabilities.

If we replace the projective measurement operator P_j with a positive operator Π_j that may not necessarily be a projection, the probability of obtaining outcome j is given by

$$p_j = \mathrm{tr}(\hat{\rho} \Pi_j) \tag{8}$$

Since Π_j is a positive operator, $p_j \geq 0$, and if we further require that the set of such operators for a finite set of outcomes is complete, meaning $\sum_j \Pi_j = \mathbb{I}$, then the quantities p_j can be interpreted as probabilities for the outcomes of the measurement described by

the operator set $\{\Pi_j\}$. Such a set of measurement operators is called a positive operator valued measure, or POVM. A key feature of POVMs is that their elements do not need to be orthogonal. Consequently, the number of possible outcomes can be larger than the dimension of the Hilbert space of the state being measured, offering greater flexibility in measurement design.

Using POVMs, we can replace the postulates of the quantum theory of standard measurements with the following more flexible ones:

1. Measurement operators are positive operators $\Pi_j \geq 0$ where the set of all such operators is complete: $\sum_j \Pi_j = \mathbb{I}$. The set of such operators is called a POVM. The elements of the POVM can be written in terms of generalized measurement operators A_j, where
$$\Pi_j = A_j^\dagger A_j \tag{9}$$
The generalized measurement operators can be non-Hermitian, though the elements of the POVM must be Hermitian by construction.

2. Each element of the POVM set represents a possible outcome of the measurement.

3. The state of the system after the measurement is
$$\hat{\rho}_j = \frac{A_j \hat{\rho} A_j^\dagger}{\mathrm{tr}(A_j \hat{\rho} A_j^\dagger)} \tag{10}$$

4. The probability that the outcome j (indexing the measurement operator Π_j) is found is
$$p_j = \mathrm{tr}(\Pi_j \hat{\rho}) \tag{11}$$

5. If we perform the measurement but do not record the results, the post-measurement state is described by the density operator
$$\hat{\rho}' = \sum_j A_j \hat{\rho} A_j^\dagger \tag{12}$$

The POVM formalism described above provides a way to describe classical information obtained from a quantum state by way of a measurement. State discrimination is the process of processing this information to determine, to the best of our knowledge, which quantum state we possess. The fundamental question of how well we can discriminate between quantum states lies at the heart of quantum mechanics. When the set of possible states are orthogonal to one another, perfect discrimination is always possible. This can be achieved by choosing a set of projection operators that correspond exactly to the

set of possible states. However, when the possible states are non-orthogonal, perfect discrimination is impossible. In such cases, we must define optimal discrimination with respect to some figure of merit, though finding this optimal discrimination is often non-trivial. The chapter will next set up the problem of state discrimination and describe two commonly used strategies: minimum-error discrimination and unambiguous discrimination. Following that, some interesting well-known measurements that do not fall into these two categories, such as the pretty-good measurement and the no-measurement strategy, will be discussed.

1.3 State Discrimination

Consider an ensemble of quantum systems such that each system is prepared in one of N possible states, say $\hat{\rho}_1, \ldots, \hat{\rho}_N$, with a priori probabilities $p_1, \ldots, p_N$ respectively, where $\sum_{j=1}^{N} p_j = 1$. In general, the states can be pure or mixed, and they can be either discrete variable states or continuous variable states. The observer performing state discrimination has full knowledge of the ensemble being prepared, meaning he knows the possible states and their prior probabilities. However, he does not know which one of the N states he possesses, and he must make a measurement to determine this. State discrimination describes how he should design and perform these measurements to optimally identify the unknown state.

There are two commonly used strategies for state discrimination. The first is discrimination with minimum error, attributed to Helstrom, who studied distinguishing between two given quantum states. Helstrom found an optimum measurement strategy for both mixed and pure states that minimizes the probability of error. This optimum measurement strategy is known as the Helstrom measurement, and the resulting minimum error is called the Helstrom bound. The general problem of discriminating between $N > 2$ states with minimum error is considerably more difficult. Although the necessary and sufficient conditions for an optimal measurement have been found, how to implement such an optimal measurement remains an open problem, except for a few special cases.

The second strategy is unambiguous discrimination, first proposed by Ivanovic. This strategy allows perfect discrimination between two non-orthogonal states, but with the caveat that a non-conclusive (inconclusive) answer is permitted. The specific case of discriminating between two pure non-orthogonal states was solved by Dieks and Peres. However, the general case of unambiguously discriminating between N pure states remains an open problem. It has been shown by Chefles that unambiguous discrimination is only possible if the states to be distinguished are linearly independent. Additionally, there are active research areas focused on extending unambiguous discrimination strategies to

mixed states.

Having described the two strategies briefly, let us now discuss each of them in more detail.

1.3.1 State Discrimination with Minimum Error

In the minimum error discrimination strategy, we always reach a conclusion about which state was prepared, though errors in this conclusion are inevitable. The goal of the optimal strategy is to minimize the detection error. To formulate this as a definite optimization problem, we must define what constitutes a detection error.

Let Π_j be the POVM element corresponding to the state $\hat{\rho}_j$ being prepared. According to the postulates of quantum measurement theory, if the actual prepared state is $\hat{\rho}$, the probability of guessing that the state is $\hat{\rho}_j$ is $\mathrm{tr}(\hat{\rho}\Pi_j)$. Since the state $\hat{\rho}_j$ occurs with a priori probability p_j, the overall probability of a correct guess is given by

$$P_{\text{corr}} = \sum_{j=1}^{N} p_j \,\mathrm{tr}(\hat{\rho}_j \Pi_j) \tag{13}$$

The detection error is then defined as

$$P_{\text{err}} = 1 - P_{\text{corr}} \tag{14}$$

To minimize the error, we must minimize P_{err} over all possible sets of POVM elements $\{\Pi_j\}$ for $j = 1, \ldots, N$ subject to the constraints $\Pi_j \geq 0$ and $\sum_{j=1}^{N} \Pi_j = \mathbb{I}$. The POVM that achieves this minimization is called the optimal measurement, and the resulting error is the minimum error. The special case of distinguishing single copies of two mixed states will be derived later in Section 3.3.1.

1.3.2 Unambiguous State Discrimination

Unlike discrimination with minimum error, in unambiguous discrimination we are not permitted to make measurement errors. However, this strategy does not succeed 100% of the time, and this trade-off is necessary to achieve error-free discrimination.

To illustrate how unambiguous state discrimination works, we consider a simple example involving two pure states $|\psi_0\rangle$ and $|\psi_1\rangle$ occurring with equal a priori probabilities, given by

$$|\psi_0\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle \tag{15}$$

$$|\psi_1\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle \tag{16}$$

These states are non-orthogonal with overlap $\langle\psi_0|\psi_1\rangle = \cos^2\theta - \sin^2\theta = \cos(2\theta)$.

A POVM for distinguishing between these states can be constructed with elements

$$\Pi_0 = a_0(\sin\theta|0\rangle + \cos\theta|1\rangle)(\sin\theta|0\rangle + \cos\theta|1\rangle) \tag{17}$$

$$\Pi_1 = a_1(\sin\theta|0\rangle - \cos\theta|1\rangle)(\sin\theta|0\rangle - \cos\theta|1\rangle) \tag{18}$$

$$\Pi_2 = \mathbb{I} - \Pi_0 - \Pi_1 \tag{19}$$

where the coefficients are chosen such that

$$\langle\psi_0|\Pi_1|\psi_0\rangle = \langle\psi_1|\Pi_0|\psi_1\rangle = 0 \tag{20}$$

and $0 \leq a_0, a_1 \leq 1$. When outcome 0 is obtained, we can confidently conclude the state is $|\psi_0\rangle$; when outcome 1 is obtained, we can confidently conclude the state is $|\psi_1\rangle$. However, if outcome 2 occurs, no definite conclusion can be made.

To evaluate how effective this unambiguous discrimination strategy is, we need to determine the probability that the measurement yields an inconclusive result. A lower probability of inconclusive outcomes indicates a better discrimination scheme. Therefore, we now derive this probability and then optimize it over the available parameters. We begin by computing the conclusive probabilities. we first compute the conclusive probabilities. The probability of obtaining outcome 0 when $|\psi_0\rangle$ is sent is

$$\langle\psi_0|\Pi_0|\psi_0\rangle = a_0|\langle\psi_0|(\sin\theta|0\rangle + \cos\theta|1\rangle)|^2 \tag{21}$$

The overlap is

$$\langle\psi_0|(\sin\theta|0\rangle + \cos\theta|1\rangle) = \cos\theta\sin\theta + \sin\theta\cos\theta = \sin(2\theta) \tag{22}$$

Thus,

$$\langle\psi_0|\Pi_0|\psi_0\rangle = a_0\sin^2(2\theta) \tag{23}$$

Similarly, for $|\psi_1\rangle$,

$$\langle\psi_1|\Pi_1|\psi_1\rangle = a_1\sin^2(2\theta) \tag{24}$$

Since the prior probabilities are equal ($p_0 = p_1 = 1/2$), the overall probability of a conclusive result is

$$P_{\text{conc}} = \frac{1}{2}a_0 \sin^2(2\theta) + \frac{1}{2}a_1 \sin^2(2\theta) \tag{25}$$

Completeness of the POVM and positivity of Π_2 imposes constraints on a_0 and a_1. The optimal choice that maximizes the conclusive probability (or equivalently minimizes the inconclusive probability) is found to be $a_0 = a_1 = 1/(1 + \cos(2\theta))$. Substituting this value, we obtain

$$\langle\psi_0|\Pi_0|\psi_0\rangle = \frac{\sin^2(2\theta)}{1 + \cos(2\theta)} = \frac{1 - \cos^2(2\theta)}{1 + \cos(2\theta)} = 1 - \cos(2\theta) \tag{26}$$

Thus, the probability of obtaining an inconclusive result when $|\psi_0\rangle$ is sent is $1 - (1 - \cos(2\theta)) = \cos(2\theta)$. The same holds for $|\psi_1\rangle$. Therefore, the overall probability of an inconclusive result is

$$P(\text{inconclusive result}) = \cos(2\theta) \tag{27}$$

This minimum value is known as the Ivanovic-Dieks-Peres (IDP) limit. Analytical solutions for the IDP limit are not known for the general case of $N > 2$ pure states, though the solution for three pure states has been derived. Various numerical techniques have also been developed to calculate the IDP limit in the general case.

1.3.3 Pretty-Good Measurement

More often than not, in minimum-error state discrimination, it is not possible to find the optimal measurement. In such cases, the pretty-good measurement (PGM), also known as the square-root measurement, is—as its name suggests—a good measurement strategy to pursue. For a given set of N possible states, we can always construct the POVM elements of the PGM as

$$\hat{\Pi}_i = p_i \hat{\rho}^{-1/2} \hat{\rho}_i \hat{\rho}^{-1/2} \text{ for } i = 1, \ldots, N \tag{28}$$

where

$$\hat{\rho} = \sum_i p_i \hat{\rho}_i \tag{29}$$

and p_i is the probability of the ith state $\hat{\rho}_i$ occurring. The POVM elements defined in this way satisfy both positivity and completeness. Although the PGM is not the optimal measurement in the general case, it turns out to be optimal for many of the cases in which the minimum-error measurement is known. These results have been summarized in the literature.

1.3.4 Other Measurement Strategies

An interesting state discrimination strategy is the no-measurement strategy. For some sets of states, the optimal measurement is to make no measurements at all and simply guess the state that is most likely. A practical example where the no-measurement strategy applies is distinguishing a state from random noise: if the signal-to-noise ratio is sufficiently low, one should always guess that the received state was random noise.

Additionally, there exist mixed strategies that allow inconclusive results while minimizing the probability of detection error, as considered by Chefles and Barnett. The suggested potential use of such mixed strategies is to provide added flexibility for an eavesdropper to "listen in" on a cryptographic protocol.

Having discussed some of the strategies for state discrimination, important questions still remain: How does state discrimination help our understanding of quantum mechanics, and how can it be used in quantum protocols? The next section will address these questions.

1.4 The Role of State Discrimination in Quantum Information Theory

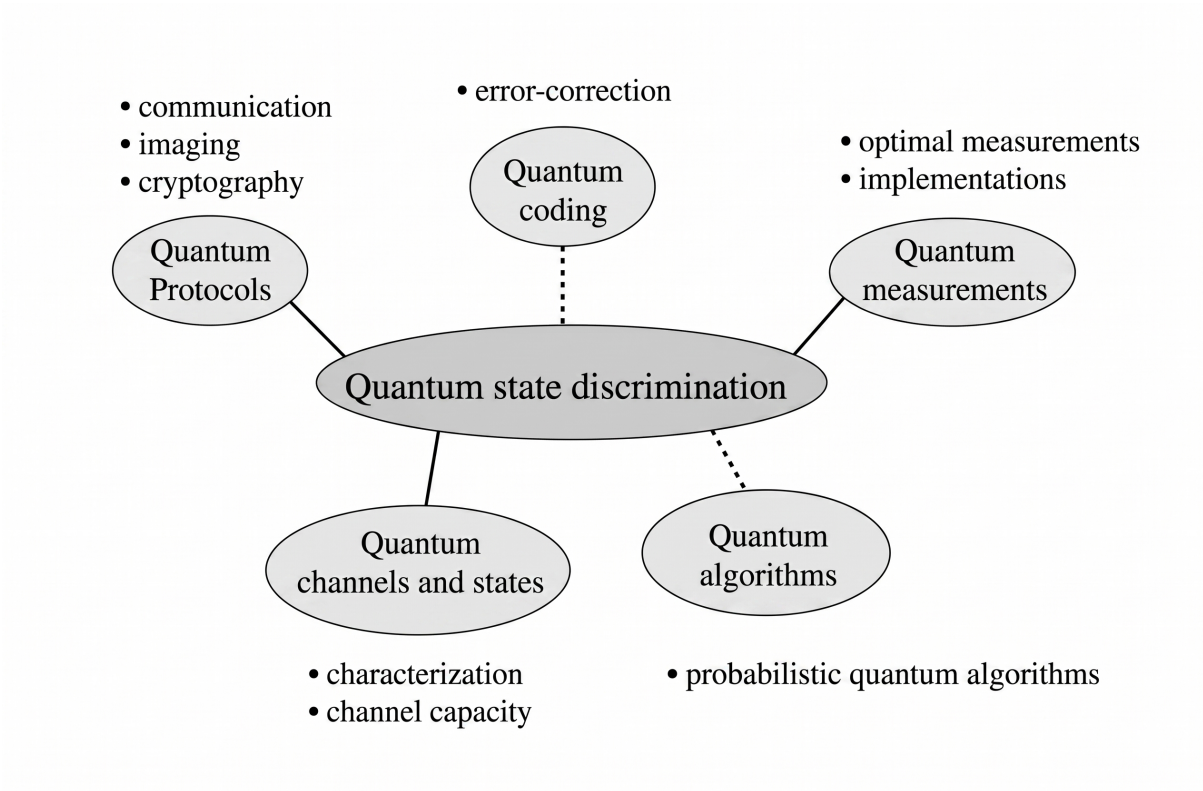

Figure 1: An overview of the central role of quantum state discrimination in quantum information theory.

Quantum state discrimination lies at the heart of quantum computation and communication. The specific strategy of state discrimination applied depends on the task in quantum information theory to be solved. For example, using state discrimination techniques, we can show that quantum mechanics does not permit perfect discrimination between nonorthogonal states $|\psi_0\rangle$ and $|\psi_1\rangle$—a result that is also directly implied by the no-cloning theorem.

To demonstrate this, assume we have a POVM with two elements Π_0 and Π_1. For perfect discrimination, we require that

$$\Pi_0|\psi_1\rangle = 0 \tag{30}$$

$$\Pi_1|\psi_0\rangle = 0 \tag{31}$$

This means that if outcome 0 is obtained, we deduce the state was $|\psi_0\rangle$, and if outcome 1 is obtained, we deduce the state was $|\psi_1\rangle$. The POVM operators must also satisfy the completeness relation

$$\Pi_0 + \Pi_1 = \mathbb{I} \tag{32}$$

To derive the necessary condition for perfect discrimination, we multiply the above equation on the left by $\langle\psi_0|$ and on the right by $|\psi_1\rangle$:

$$\langle\psi_0|(\Pi_0 + \Pi_1)|\psi_1\rangle = \langle\psi_0|\mathbb{I}|\psi_1\rangle \tag{33}$$

Distributing the left-hand side gives

$$\langle\psi_0|\Pi_0|\psi_1\rangle + \langle\psi_0|\Pi_1|\psi_1\rangle = \langle\psi_0|\psi_1\rangle \tag{34}$$

Now, from the perfect discrimination conditions, we have $\Pi_0|\psi_1\rangle = 0$. Therefore, the first term $\langle\psi_0|\Pi_0|\psi_1\rangle = 0$. For the second term, note that Π_1 is a positive operator. If $\Pi_1|\psi_0\rangle = 0$, then taking the adjoint gives $\langle\psi_0|\Pi_1 = 0$. However, this does not directly imply $\langle\psi_0|\Pi_1|\psi_1\rangle = 0$. In standard unambiguous discrimination, the POVM elements are constructed such that Π_1 projects onto a subspace orthogonal to $|\psi_0\rangle$, which indeed forces $\langle\psi_0|\Pi_1|\psi_1\rangle = 0$. Under this construction, we have,

$$0 + 0 = \langle\psi_0|\psi_1\rangle \tag{35}$$

Thus,

$$\langle\psi_0|\psi_1\rangle = 0 \tag{1.20}$$

This condition can only be satisfied if the states $|\psi_0\rangle$ and $|\psi_1\rangle$ are orthogonal. Conse-

quently, perfect discrimination is impossible for non-orthogonal states. Insights such as these are prevalent when applying state discrimination to problems in quantum information theory. The fact that we cannot perfectly distinguish between non-orthogonal states is what makes quantum key distribution possible.

Another important use of state discrimination is to characterize properties of quantum states, such as entanglement and separability. Much research has been devoted to this area. One of the main motivations behind such studies is to determine whether entangled states are useful for certain tasks, or whether entanglement is needed for a given task at all. Quantum metrology, quantum imaging, and the classical capacity of quantum channels are examples of such investigations.

State discrimination is also indispensable in quantum error correction. Error correction cannot be performed without syndrome measurement. A syndrome measurement determines whether a quantum state has been corrupted and, if so, which part of the state has been affected. The syndrome measurement is typically a multi-state measurement that does not disturb the encoded state but still provides information about the error. Such measurements lie at the heart of error correction.

Finally, state discrimination has found uses in quantum algorithms through a variant known as quantum state filtering. In quantum state filtering, the goal is to determine whether an unknown quantum state is a specific state $|\psi_k\rangle$ from a set of known states $\{|\psi_1\rangle, \ldots, |\psi_N\rangle\}$, or one of the others. Using optimal solutions in quantum state filtering, it is possible to implement an efficient probabilistic quantum algorithm that distinguishes between sets of Boolean functions, which generalizes the Deutsch-Jozsa algorithm. It has also been proposed that quantum decision problems can be characterized by state discrimination.

Having discussed the role of state discrimination in the major topics of quantum information theory, we now introduce the encoding of information onto continuous variable states.

1.5 Continuous Variables in Quantum Mechanics

While quantum information is primarily concerned with discrete quantum variables, such as qubits, continuous variable states have always played an important role in quantum mechanics. The position and momentum of the quantized harmonic oscillator are continuous variables. Because a mode of the electromagnetic field can be thought of as a harmonic oscillator, its quadrature amplitudes (the electric and magnetic fields) are continuous

variables. Due to this connection to the quantized electromagnetic field, the development of the laser in the 1960s inspired considerable research into the properties of continuous quantum variables.

Continuous variables also play an important role in quantum information. The original EPR thought experiment was formulated in terms of the entangled momentum and position states of two particles. Furthermore, one of the first experimental demonstrations of quantum teleportation was performed using continuous variable states. Many aspects of discrete quantum information theory have continuous analogues, including quantum codes. It is even possible to formulate a theory of universal quantum computation over continuous variables by combining linear optical devices such as phase shifters and beamsplitters with nonlinear elements such as downconverters and parametric amplifiers.

The primary reason for developing a theory of quantum information over continuous variables is a practical one. Most communication channels involve the exchange of bosons (such as photons or phonons), which correspond to quantized harmonic oscillators. Accordingly, the theory of quantum information over continuous variables is essential for understanding the physical limits of communication. Moreover, many important sensing and measurement techniques, including microscopy and radar, rely on bosonic systems. Quantum information over continuous variables is the key to understanding the limits of sensing and measurement and for devising methods to attain those limits. With this motivation, this work explores the fundamental limits of quantum communication and sensing and introduces techniques to achieve them, beginning with a review of continuous variable quantum information theory in the following chapter.

1.6 Scope of This Work

This subsection lists the major research areas in continuous variable quantum computing and quantum information, and situate the contributions of this work within them.

Major areas of research include:

1. **Channel Capacity of the Noisy, Lossy Bosonic Channel:** A fundamental question in quantum communication is: what is the maximum rate at which information can be sent down a bosonic channel in the presence of loss and noise? The capacity of a quantum channel $\mathcal{E}$ with inputs drawn from an ensemble of states $\hat{\rho}_k$ with probabilities p_k is given by its Holevo quantity,

$$\chi(\{p_k, \hat{\rho}_k\}) = S\left(\sum_k p_k \mathcal{E}(\hat{\rho}_k)\right) - \sum_k p_k S\left(\mathcal{E}(\hat{\rho}_k)\right), \tag{36}$$

where $S(\hat{\rho})$ is the von Neumann entropy of the state $\hat{\rho}$. Maximizing this quantity over all possible ensembles, including entangled inputs across multiple uses of the channel, yields the channel capacity. This maximization is often difficult to perform. The capacity of the bosonic channel with linear loss is known and is attained by sending ensembles of coherent states down the channel. By contrast, the capacity of the bosonic channel with linear loss and thermal noise remains unknown, although it is conjectured to also be attained by sending ensembles of coherent states.

2. **Minimum Output Entropy Conjecture:** The capacity of the bosonic channel with linear loss and thermal noise is indeed attained by coherent state inputs if the bosonic minimum output entropy conjecture holds. This conjecture states that a coherent-state input yields the output with the minimum von Neumann entropy for a bosonic channel with additive Gaussian noise. Proving or disproving the bosonic minimum output entropy conjecture is perhaps the most important open problem in continuous variable quantum information theory.

3. **Optimal Measurements for Distinguishing Bosonic States:** Distinguishing quantum states is an important component of quantum information science. Key results, such as the no-cloning theorem, the quantum teleportation protocol, and more recently the Yuen no-go theorem, have their origins in problems of state discrimination. The Helstrom bound gives the minimum probability of error technique for discriminating between two quantum states. The quantum Chernoff bound is an upper bound to the Helstrom limit of multiple-copy binary state discrimination, which is asymptotically tight in the error exponent. While the measurement techniques for attaining the Helstrom bound are known in principle, devising experimentally practical measurement techniques for attaining the Helstrom bound is not known in general. Moreover, even when the channel capacity for a quantum channel is attained by product state (unentangled) inputs, the measurement required to attain the capacity is the entangling 'pretty-good' measurement, which may be hard to perform in practice. For example, even though the channel capacity for the bosonic channel with linear loss is attained by coherent state inputs, practical decoding techniques that attain this bound are not currently known.

2 Continuous Variable Quantum Information Theory

3 State Discrimination Using Gaussian States

3.1 Introduction

With the basic formalism and notation for continuous variable states established, we now address the problem of discriminating between such states, focusing particularly on

Gaussian states.

A continuous variable state is classified as Gaussian if its Wigner function – or equivalently, its characteristic function – takes a Gaussian form. Mathematically, this is expressed as:

$$W[\hat{\rho}](\vec{R}) = \frac{1}{(2\pi)^n\sqrt{\det\sigma}} \exp\left(-\frac{1}{2}(\vec{R}-\bar{R})^T\sigma^{-1}(\vec{R}-\bar{R})\right) \tag{37}$$

where,

- $W[\hat{\rho}](\vec{R})$: The Wigner function tells us the 'probability-like' value at point $\vec{R}$ in phase space.
- $\vec{R}$: This is a vector containing the positions and momenta of all n modes. For example, for one mode, $\vec{R} = (q, p)$.
- $\bar{R}$: The average (mean) value of $\vec{R}$. This is the center of the bell curve.
- σ: The covariance matrix. It tells us how spread out the state is in different directions.
- $\det\sigma$: The determinant of the covariance matrix. It measures the overall volume of the distribution.
- n: The number of modes (like the number of particles or light beams).

Alternatively,

$$\chi[\hat{\rho}](\vec{\Lambda}) = \exp\left(-\frac{1}{2}\vec{\Lambda}^T\sigma\vec{\Lambda} + i\vec{\Lambda}^T\bar{R}\right) \tag{38}$$

where,

- $\chi[\hat{\rho}](\vec{\Lambda})$: The characteristic function. It is like the 'moment-generating' function in statistics.
- $\vec{\Lambda}$: A vector in phase space that is conjugate to $\vec{R}$ (like frequency is conjugate to time).
- σ: The same covariance matrix from Equation (3.1).
- $\bar{R}$: The same mean vector from Equation (3.1).

Thus, a Gaussian state $\hat{\rho}$ is completely characterized by its covariance matrix σ and its mean vector $\bar{R}$. The covariance matrix gains further physical significance through Williamson's Theorem, which decomposes it into a symplectic matrix and a diagonal

matrix of symplectic eigenvalues. Importantly, all entanglement properties of a Gaussian state are contained within σ. This is because local operations do not affect entanglement, so the mean vector $\bar{R}$ can be arbitrarily adjusted through phase space translations without altering the entanglement of $\hat{\rho}$. Conversely, generating entanglement necessarily requires a nonlocal operation on $\hat{\rho}$.

3.2 Normal form decomposition

A useful theorem for expressing the covariance matrix of a Gaussian state is Williamson's Theorem, stated as follows:

Theorem 3.1 (Williamson's Theorem)**.** *Given a real $2n \times 2n$ symmetric and positive matrix σ, there exists a real and symplectic $2n \times 2n$ matrix S and a real $n \times n$ diagonal and positive matrix W such that*

$$\sigma = S^T W S \tag{39}$$

where

$$W = \bigoplus_{k=1}^{n} d_k \mathbb{I}_2 \tag{40}$$

and

$$\mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{41}$$

Proof. The proof proceeds in several steps. We will construct the symplectic matrix S that diagonalizes σ in the symplectic sense.

Since σ is real, symmetric, and positive-definite, it can be diagonalized by an orthogonal matrix $O \in O(2n)$:

$$\sigma = O^T D O, \tag{42}$$

where $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_{2n})$ with $\lambda_i > 0$. However, this is an orthogonal diagonalization, not a symplectic one. Our goal is to find a symplectic matrix S that brings σ to the block-diagonal form W.

Define the matrix

$$M = i\Omega\sigma, \tag{43}$$

where Ω is the symplectic form:

$$\Omega = \bigoplus_{k=1}^{n} \omega, \qquad \omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{44}$$

Consider the matrix $M = i\Omega\sigma$. Its transpose is:

$$M^T = (i\Omega\sigma)^T = i\sigma^T\Omega^T = i\sigma(-\Omega) = -i\sigma\Omega. \tag{45}$$

Now note that:

$$\Omega^{-1}M\Omega = \Omega^{-1}(i\Omega\sigma)\Omega = i\sigma\Omega. \tag{46}$$

Since $\sigma\Omega$ is similar to $-\Omega\sigma$ (because $\Omega^T = -\Omega$), the eigenvalues of M come in $\pm$ pairs. With the factor i, the eigenvalues are of the form $\pm id_k$ where $d_k > 0$.

Let the eigenvalues of M be $\pm id_1, \pm id_2, \ldots, \pm id_n$ with $d_k > 0$. For each eigenvalue id_k, there exists a complex eigenvector u_k such that:

$$i\Omega\sigma u_k = id_k u_k \quad \Rightarrow \quad \Omega\sigma u_k = d_k u_k. \tag{47}$$

Similarly, for eigenvalue $-id_k$, we have an eigenvector v_k satisfying $\Omega\sigma v_k = -d_k v_k$.

Using the fact that σ is symmetric and Ω is skew-symmetric, one can show that v_k is proportional to the complex conjugate of u_k. We choose them such that:

$$v_k = i\sigma u_k. \tag{48}$$

This choice ensures the correct normalization.

We normalize the eigenvectors such that:

$$u_k^\dagger \sigma u_k = 1. \tag{49}$$

Then, using properties of Ω and σ, we also get:

$$u_k^\dagger \Omega u_k = i. \tag{50}$$

Define the real vectors x_k and y_k for $k = 1, \ldots, n$ as:

$$x_k = \sqrt{2}\ \mathrm{Re}(u_k), \qquad y_k = \sqrt{2}\ \mathrm{Im}(u_k). \tag{51}$$

One can verify that these vectors satisfy:

$$x_k^T \sigma x_k = y_k^T \sigma y_k = d_k, \tag{52}$$

$$x_k^T \sigma y_k = 0, \tag{53}$$

and

$$x_k^T \Omega x_k = y_k^T \Omega y_k = 0, \tag{54}$$

$$x_k^T \Omega y_k = 1. \tag{55}$$

Let S be the $2n \times 2n$ matrix whose columns are, in order:

$$S = \begin{pmatrix} x_1 & y_1 & x_2 & y_2 & \cdots & x_n & y_n \end{pmatrix}. \tag{56}$$

From the properties above:

- Symplectic condition: $S^T \Omega S = \Omega$ (since $x_k^T \Omega y_k = 1$ and all other pairings are zero).
- Diagonalization of σ: $S^T \sigma S = \bigoplus_{k=1}^n d_k \mathbb{I}_2 = W$.

Thus, we have constructed a symplectic matrix S such that:

$$S^T \sigma S = W \quad \Rightarrow \quad \sigma = (S^{-1})^T W S^{-1}. \tag{57}$$

Since S is symplectic, S^{-1} is also symplectic. Renaming S^{-1} as S (or absorbing the inverse into the definition), we obtain:

$$\sigma = S^T W S \tag{58}$$

with $S \in Sp(2n, \mathbb{R})$ and W diagonal as required. This completes the proof. □

The elements d_k of the matrix W are called the symplectic eigenvalues and can be calculated from the spectrum of $i\Omega\sigma$, while the matrix S is the symplectic matrix that performs the symplectic decomposition. Finding the symplectic eigenvalues of a covariance matrix is usually straightforward, but determining the symplectic matrix that performs the decomposition is often difficult. Performing the symplectic decomposition of Gaussian states is a key part of the process of determining the distinguishability of those states. For example, it turns out that in order to calculate the quantum Chernoff bound for Gaussian states – an asymptotic distinguishability measure for states that will be used extensively – we must find the symplectic decomposition of the states involved.

The phenomenological statement implied by equation (3.3) is that every Gaussian state $\hat{\rho}_G$ can be obtained from a thermal state $\hat{\rho}_{\text{th}}$ (described by a diagonal covariance matrix as in equation (2.96)) by performing the unitary transformation U_S associated with the symplectic matrix S that performs the symplectic decomposition:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{59}$$

where $\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \hat{\rho}_{\text{th},2} \otimes \cdots \otimes \hat{\rho}_{\text{th},n}$ is a product of thermal states $\hat{\rho}_{\text{th},k}$ for $k = 1, \ldots, n$. Now, we derive the above equation.

From Williamson's Theorem, we have:

$$\sigma = S^T W S \tag{3.3}$$

where:

- σ is the covariance matrix of an arbitrary Gaussian state $\hat{\rho}_G$
- S is a symplectic matrix ($S \in Sp(2n, \mathbb{R})$)
- W is a diagonal matrix of the form:

$$W = \bigoplus_{k=1}^{n} d_k \mathbb{I}_2, \qquad \mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{60}$$

For a single-mode thermal state $\hat{\rho}_{\text{th},k}$ with average photon number N_k, the covariance matrix is (from equation 2.96):

$$\sigma_{\text{th},k} = \frac{1}{4} \begin{pmatrix} 2N_k + 1 & 0 \\ 0 & 2N_k + 1 \end{pmatrix} = \left(N_k + \frac{1}{2} \right) \mathbb{I}_2 \tag{61}$$

Comparing this with the block form of W, we identify:

$$d_k = N_k + \frac{1}{2} \tag{62}$$

Thus, the matrix W can be written as:

$$W = \bigoplus_{k=1}^{n} \left(N_k + \frac{1}{2} \right) \mathbb{I}_2 \tag{63}$$

This is exactly the covariance matrix of a **product of thermal states**:

$$\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \hat{\rho}_{\text{th},2} \otimes \cdots \otimes \hat{\rho}_{\text{th},n} \tag{64}$$

whose covariance matrix is:

$$\sigma_{\text{th}} = W \tag{65}$$

Now, we want to find a unitary transformation U_S that maps the thermal state $\hat{\rho}_{\text{th}}$ to the target Gaussian state $\hat{\rho}_G$.

In phase space, a unitary transformation generated by a quadratic Hamiltonian corresponds to a symplectic transformation on the covariance matrix (see Appendix C of

the thesis):

$$\sigma \to S\sigma S^T \tag{66}$$

More precisely, if $\hat{\rho}' = U_S \hat{\rho} U_S^\dagger$, then the covariance matrix transforms as:

$$\sigma' = S\sigma S^T \tag{67}$$

where S is the symplectic matrix associated with the unitary U_S.

Let us apply the symplectic transformation S to the thermal state $\hat{\rho}_{\text{th}}$:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{68}$$

The covariance matrix of $\hat{\rho}_G$ becomes:

$$\sigma_G = S\sigma_{\text{th}} S^T \tag{69}$$

But $\sigma_{\text{th}} = W$, so:

$$\sigma_G = SWS^T \tag{70}$$

Williamson's Theorem states:

$$\sigma = S^T W S \tag{71}$$

Notice the difference: Williamson's theorem has $S^T W S$, while we have SWS^T. However, since S is symplectic, its inverse is given by:

$$S^{-1} = \Omega S^T \Omega^{-1} \tag{72}$$

More simply, if S is symplectic, then S^T is also symplectic. Therefore, we can rename S^T as a new symplectic matrix S'. This means:

$$SWS^T = (S^T)^T W (S^T) = S'^T W S' \tag{73}$$

Thus, both forms are equivalent. Williamson's theorem guarantees the existence of some symplectic matrix such that $\sigma = S^T W S$. Therefore, there exists a unitary transformation U_S such that:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{74}$$

Thus, we arrive at the equation:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{75}$$

where:

- $\hat{\rho}_G$ is any Gaussian state
- $\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \cdots \otimes \hat{\rho}_{\text{th},n}$ is a product of thermal states
- U_S is the unitary transformation associated with the symplectic matrix S

3.3 Distinguisability Of States

Although, the problem of distinguisability is defined for any number of states, let us start with the problem of distinguisability of two states, $\hat{\rho}_0$ and $\hat{\rho}_1$, with their corresponding equiprobable binary hypothesis H_0 and H_1 respectively. However, so far, we have only discussed more of the formalism, but nothing that could be used quantify the distinction between two states. In particular, we want something that can quantify some *distance* between the two states or some kind of method to quantify some *error* for distinction between two states given the measurement. For the notion of *distance*, we use *fidelity of states* and as discused in chapter 1, the *Helstrom's Bound* can be viewed as a measure of the error which gives the minimum probability of error and the corresponding POVM that gives the minimum measurement errors.

3.3.1 Helstrom's Optimal Discrimination

As discussed in chapter 1, Helstrom's Bounds corresponds to the generic optimization problem to find the POVM that gives the best measurement with minimum error probability. Given the two states, we consider a dichotomous POVM $\{\hat{E}_0, \hat{E}_1\}$, where $\hat{E}_0$ corresponds to the hypothesis $\hat{H}_0$ and $\hat{E}_1$ corresponds to the hypothesis $\hat{E}_1$ respectively. The error probability can simply be obtained from the hypothesis testing for H_0 and H_1 as

$$P_E = \frac{1}{2}(P(H_0|H_1) + P(H_1|H_0)) \tag{76}$$

where,
$P(H_0|H_1)$ is the probability of concluding that H_0 is true, when H_1 is and
$P(H_1|H_0)$ is the probability of concluding that H_1 is true, when H_0 is true and
the factor of $\frac{1}{2}$ arises from the equiprobable assumptions.

Also, note that the expectation of an operator $\hat{O}$ given the system is in state $\hat{\rho}$ is given by $\left\langle \hat{O} \right\rangle = tr(r\hat{h}o\hat{O})$. So, if the system is in $\hat{\rho}_j$ and we make a measurement with the operator with $\hat{E}_i$, then the probability of error in the measurement is given by

$$P(H_i|H_j) = tr(\hat{\rho}_j \hat{E}_i) \tag{77}$$

Thus, using eq 77 in eq 76, we get

$$P_E = \frac{1}{2}(tr(\hat{\rho}_1\hat{E}_0) + tr(\hat{\rho}_0\hat{E}_1)) \tag{78}$$

Also, since we defined our POVM to be the set $\hat{E}_0, \hat{E}_1$ and POVM are complete, so $\hat{E}_0 + \hat{E}_1 = I \implies \hat{E}_0 = I - \hat{E}_1$. So, eq 78 will be,

$$\begin{aligned} P_E &= \frac{1}{2}(tr(\hat{\rho}_1(I - \hat{E}_1)) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(tr(\hat{\rho}_1) - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(1 - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)), \text{since } tr(\hat{\rho}_i) = 1 \\ &= \frac{1}{2}(1 - tr((\hat{\rho}_1 - \hat{\rho}_0)\hat{E}_1)) \end{aligned} \tag{79}$$

Note, all we aim to is find the correct POVM set so that the error probability is minimum. For this case, as $\hat{E}_0 + \hat{E}_1 = I$, so, we can express $\hat{E}_0$ in terms of $\hat{E}_1$. Thus, we have to minimize P_E over the measurement operator $\hat{E}_1$. Define $\hat{\gamma} = \hat{\rho}_1 - \hat{\rho}_0$. Clearly, $tr(\hat{\gamma}) = tr(\hat{\rho}_1) - tr(\hat{\rho}_0) = 1 - 1 = 0$. Since, both $\hat{\rho}_0$ and $\hat{\rho}_1$ are both positive and semi definite, so the eigenvalues of $\hat{\gamma}$ can be both positive or negative. Since both $\hat{\rho}_0$ and $\hat{\rho}_1$ are hermitian, so $\hat{\gamma}$ is also hermitian, hence it has the spectral decomposition,

$$\hat{\gamma} = \sum_k \mu_k |k\rangle \langle k| \tag{80}$$

where μ_k are the real eigenvalues. Now expanding $\hat{E}_1$ in the same basis

$$\hat{E}_1 = \sum_{k,l} e_{kl} |k\rangle \langle l| \tag{81}$$

Hence,

$$\begin{aligned} tr(\hat{E}_1\hat{\gamma}) &= tr\left(\sum_{k,l} e_{kl} |k\rangle \langle l| \cdot \sum_m \mu_m |m\rangle \langle m|\right) \\ &= \sum_{klm} e_{kl}\mu_m \langle l|m\rangle \langle m|k\rangle \\ &= \sum_{klm} e_{kl}\mu_m \delta_{l,m}\delta_{m,k} \\ &= \sum_k e_{kk}\mu_k \end{aligned} \tag{82}$$

Since POVMs are positive valued, so e_{kk} is positive. Now, μ_k can be positive or negative, but since we are optimizing e_{kk}, if μ_k is negative, it will lead to negative propagation

of the errors, thus for such cases e_{kk} must be 0. Hence the net contribution will come only from the positive e_{kk} values. Thus the optimal $\hat{E}_1$ is one that projects onto positive part of $\hat{\gamma}$, $\hat{\gamma}_+$ (say) which projects to the subspace by the eigenstates of $\hat{\gamma}$ with positive eigenvalues. Thus, we have $tr(\hat{\gamma}\hat{E}_1) = tr(\hat{\gamma}_+) = \frac{1}{2}||\hat{\gamma}||_1$ where $||\hat{\gamma}||_1 = tr(|\hat{\gamma}|) = tr(\sqrt{\gamma^\dagger\gamma})$. Thus, using this, eq 79 can be written as

$$P_E^{min} = \frac{1}{2}(1 - ||\hat{\rho}_1 - \hat{\rho}_0||_1) \tag{83}$$

However, the calculation of the minimum probability of error using the above hypothesis is only for a single particle system with two states. But, the same analysis and calculations for an N particle system would be very difficult. This is because as the number of particles in the system increases the Hilbert Space grows exponentially over N as the N copy density matrix resides in the tensor product state of dimension d^N, where d is the dimension (degrees of freedom) for a single particle state, which may not simplify for larger N values. Moreover, opimization of such d^N matrix set (POVM) would also be analytically as well as numerically very difficult. However, one can calculate the *Quantum Chernoff Bounds*, which gives the asymptotic dependency of N copys of the system on the probability of error. Then the quantum hypothesis test becomes

$$\begin{aligned} H_0 &: \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0 \\ H_1 &: \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 \end{aligned} \tag{84}$$

Thus for this case, the minimum probability of error is given by

$$P_{E,N}^{min} = \frac{1}{2}(1 - ||\hat{\gamma}_N||_1) \tag{85}$$

where $\hat{\gamma}_N = \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 - \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0$

3.3.2 Fidelity

Fidelity between two states could be used as a notion of measure of the *distance* between two states. For two states, $\hat{\rho}_1$ and $\hat{\rho}_2$, fidelity is defined as,

$$F(\hat{\rho}_1, \hat{\rho}_2) = \left(tr\left(\sqrt{\sqrt{\hat{\rho}_1}\hat{\rho}_2\sqrt{\hat{\rho}_1}} \right) \right)^2 \tag{86}$$

- F($\hat{\rho}_1$, $\hat{\rho}_2$) ≤ 1
- F($\hat{\rho}_1$, $\hat{\rho}_2$) = F($\hat{\rho}_2$, $\hat{\rho}_1$)
- If $\hat{\rho}_1$ is a pure state, $\hat{\rho}_1 = |\psi_1\rangle\langle\psi_1|$, then F($\hat{\rho}_1$, $\hat{\rho}_2$) = $\langle\psi_1|\hat{\rho}_2|\psi_1\rangle$
- F($\hat{\rho}_1$, $\hat{\rho}_2$) is invariant under unitary transformations on the state space.

- $\mathrm{F}(\hat{\rho}_1 \otimes \hat{\sigma}_1,\, \hat{\rho}_2 \otimes \hat{\sigma}_2) = \mathrm{F}(\hat{\rho}_1,\, \hat{\rho}_2)\ \mathrm{F}(\hat{\sigma}_1,\, \hat{\sigma}_2)$

3.3.3 Quantum Chernoff Bounds

So far, we have discussed how the error is quantified and methods of quantifying the error. But, how is the actual minimization of the errror probability occurs. Chernoff solved this problem in the asymptotic regime and gave the asymptotic dependence on the probability of error if we have N copies of the system, i.e. a N trail hypothesis testing.[2] He showed that the probability of error decreases exponentially with the number of tests performed N, i.e. $P_E \approx exp(-\kappa N)$. The optimal exponent κ arising in the asymptotic limit is called Chernoff Bounds[2]. But why is Chernoff Bounds even considered... This may go back to 'One of the virtues of the Chernoff bound is that it yields a very natural distance measure between probability distributions; it is essentially the unique distance measure in the ubiquitous situation of independent and identically distributed (i.i.d.) random variables.'[2]

The quantum chernoff bounds serves as the upper limit the the minimum probability of error 85, i.e.

$$P_{N,E}^{min} \leq P_{QC}^{N} = \frac{1}{2} exp(-\kappa N) \tag{87}$$

where $\kappa = -ln\left(\min_{\,0\leq s\leq 1} tr(\hat{\rho}_0^s \hat{\rho}_1^{(1-s)})\right)$

Before diving deep into the quantum chernoff bounds, let us discuss a bit of the classical chernoff bounds. Consider a random variable X and $a \in \mathbb{R}$. Then, [3]

$$\begin{aligned} P(X \geq a) &\leq min_{s>0} e^{-sa} M_X(s) \\ P(X \leq a) &\leq min_{s<0} e^{-sa} M_X(s) \end{aligned} \tag{88}$$

For the optimal decision, we define the log likelihood,

$$L(x) = log \frac{p_1(x)}{p_0(x)}$$

For N trial observations

$$X = \sum_{i=1}^{N} L(x_i) = \sum_{i=1}^{N} log \frac{p_1(x_i)}{p_0(x_i)}$$

where we declare H_1 is true if X ¿ 0 else we declare H_0 is true. Now we refer to the eq 76, which using above under hypothesis H_1

$$P(H_0|H_1) = P(X \leq 0|H_1) \leq e^{-s\cdot 0} M_X(s) = M_X(s)$$

where $s < 0$ The moment generating function $M_X(s)$ is defined as $\langle e^{sX} \rangle$, Hence

$$\begin{aligned} M_X(s) &= \langle e^{sX} \rangle = \langle e^{s \sum_{i=1}^{N} log \frac{p_1(x_i)}{p_0(x_i)}} \rangle \\ M_X(s) &= \langle e^{\sum_{i=1}^{N} log \frac{p_1(x_i)}{p_0(x_i)}^s} \rangle \\ M_X(s) &= \langle \prod_{i=1}^{N} e^{log \frac{p_1(x_i)}{p_0(x_i)}^s} \rangle \\ M_X(s) &= \prod_{i=1}^{N} \langle \frac{p_1(x_i)}{p_0(x_i)}^s \rangle \end{aligned} \tag{89}$$

Since, each factor is identical and $\langle f \rangle = \sum_{x_i} p(x) f(x)$ so,

$$M_X(s) = \left(\sum_x p_1(x) \left(\frac{p_1(x)}{p_0(x)} \right)^s \right)^N = \left(\sum_x p_1(x)^{1+s} p_0(x)^{-s} \right)^N \tag{90}$$

Taking $s \to -s$, i.e. $s > 0$

$$M_X(s) = \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{91}$$

$$P(H_0|H_1) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{92}$$

Similarly for $P(H_1|H_0)$, we get

$$P(H_1|H_0) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{93}$$

where $s > 0$ Combining eq 92 and eq 93 with eq 76

$$P_E^N \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{94}$$

Notice this takes the same form as the quantum Chernoff Bounds upto a factor of $\frac{1}{2}$, where instead of the probability desity functions $p_0(x)$ and $p_1(x)$, we have the density matrices ρ_0 and ρ_1.

One of the notable observation is that the Chernoff Bounds are exponentially tight. As the number of trails increases i.e. $N \to \infty$, $ln(P_{N,E}^{min})/N \to -\kappa$

Also, defining $Q_s = tr(\hat{\rho}_0^s \hat{\rho}_1^{1-s})$ and $Q = \min_{0 \leq s \leq 1} Q_s$, the quantum Chernoff Bounds can be written as

$$P_{QC}^N = \frac{1}{2} Q^N \tag{95}$$

The aim to find the minimum error would require the minimization of Q_s over all valures of $s \in [0,1]$. Setting s = $\frac{1}{2}$, we will arrive at the *Quantum Bhattacharyya Bound*, however this bound is not assymptotically tight as the Quantum Chernoff Bound.

$$P_B^N = \frac{1}{2}\left(tr(\sqrt{(}\hat{\rho}_0\hat{\rho}_1))\right)^N \tag{96}$$

If the two states are indistinguisable, i.e. $\hat{\rho}_0 = \hat{\rho}_1$, then $P_{QC}^N \approx P_B^N$ and

$$\frac{1-\sqrt{1-F(\hat{\rho}_0,\hat{\rho}_1)}}{2} \leq P^{(1)} \leq P_{QC}^{(1)} \leq \frac{\sqrt{F(\hat{\rho}_0,\hat{\rho}_1)}}{2}$$

Also, it is noted that Q_s is a convex function over s and joint-concave over $\hat{\rho}_0$ and $]r\hat{h}o_1$. Q_s can be written as

$$Q_s = tr(\hat{\rho}_0^s]r\hat{h}o_1^{1-s}) = \sum_k e_k^{(0)^s} e_k^{(1)^{1-s}}$$

where $e_k^{(0)}$ and $e_k^{(1)} \geq 0$ Thus each element of Q_s is $f_k(s) = a_k^s b_k^{1-s} = e^{slna_k} e^{(1-s)lnb_k}$, taking the 2nd derivative of f_k w.r.t. s, we see $f_k^{''}(s) = (lna_k - lnb_k)^s \cdot f_k(s) \geq 0$. Hence Q_s is convex over s. Also using Lieb's Theorem [4]

Theorem 3.2. *Lieb's Theorem: Let K be a fixed matrix in $\mathbb{C}^{n\times m}$. Then for any $t \in [0,1]$, the map*

$$(A,B) \to tr[K^* A^{1-t} K B^t]$$

is jointly concave in (A, B) where A and B are respectively $n \times n$ and $m \times m$ Hermitian positive definite matrices.

Taking $A = \hat{\rho}_0$ and $B = \hat{\rho}_1$, and $K = \mathbb{I}$ and $t = s$, then Q_s is concave over $\hat{\rho}_0$ and $\hat{\rho}_1$.

3.4 Fidelity for Gaussian States

3.4.1 Single Mode Gaussian States

For two single mode gaussian thermal states with mean 0 and covariance matrices σ_1 and σ_2 corresponding to states $\hat{\rho}_1$ and $\hat{\rho}_2$ is given by

$$F_0(\hat{\rho}_1,\hat{\rho}_2) = \frac{2}{\sqrt{det(\sigma_1+\sigma_2)}(\sqrt{det(\theta)} - \sqrt{det(\theta)-1})} \tag{97}$$

where,
$det(\theta) = 1 + \frac{P}{det(\sigma_1+\sigma_2)}$
$P = (det(\sigma_1)-1)(det(\sigma_2)-1)$
It was also derived in the paper, the fidelity for a two single mode displaced squeezed thermal states. By the Williamson's Theorem, it could be seen that any arbitrary

single mode gaussian state can be expressed as a displaced squeezed thermal state, i.e. $\hat{\rho}_G = \hat{U}_S \hat{\rho}_{th} \hat{U}_S^\dagger$. Thus, the fidelity for any single mode gaussian state,

$$F(\hat{\rho}_1, \hat{\rho}_2) = F_0(\hat{\rho}_1, \hat{\rho}_2) exp\left(-\frac{1}{2}(\overline{R_1} - \overline{R_2})^T \kappa (\overline{R_1} - \overline{R_2})\right) \tag{98}$$

where $F_0(\hat{\rho}_1, \hat{\rho}_2)$ is defined as eq 97 and $\kappa = \mathbb{I} - (\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)$

3.4.2 Multimode Gaussian States

For comparing two multi-mode gaussian states, the fidelity is defined as [5]

$$F(\hat{\rho}_1, \hat{\rho}_2) = \sqrt{Ldet(\Phi(\mathcal{O}))}$$

where,
$\Phi(A) = A(\mathbb{I} + \sqrt{\mathbb{I} + (JA)^{-1}})$
$L^{-1} = det(\Phi(\sigma_1))^{-1} det(\frac{\Phi(\sigma_1)+\sigma_2}{2}) det(\frac{\sigma_2+\Phi(\sigma_1)-U}{2})$
$U = (\sigma_1 - i\Omega)(\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)$
$\mathcal{O} = \Phi(\sigma_1) - (\Phi(\sigma_1) - i\Omega)\left(\sigma_2 + \Phi(\sigma_1) - (\sigma_2 - i\Omega)(\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)\right)^{-1}(\Phi(\sigma_1) + i\Omega)$

3.5 Quantum Chernoff Bounds for Gaussian States

Consider two n-mode Gaussian states, viz $\hat{\rho}_A$ and $\hat{\rho}_B$ with symplectic decomposition matrix S_A and S_B respectively, with symplectic eigenvalues $\{\alpha_k, k = 1, ..., n\}$ and $\{\beta_k, k = 1, ..., n\}$ respectively. The mean in the phase space are $\bar{x_A}$ and $\bar{x_B}$ respectively.

We know by Williamson's Theorem that every n-mode gaussian state ρ can be specified by a displacement vector $\bar{x} \in \mathcal{R}^{2n}$, a symplectic matrix S that preserves Ω and the symplectic eigenvalues $\nu_k, k = 1, ..., n$, where $\nu_k \geq 1$ and via the dicomposition

$$\rho = \hat{U}_{\bar{x}S} \rho_{th} \hat{U}_{\bar{x}S}^\dagger$$

where $\rho_{th} = \bigotimes_{k=1}^n \sigma\nu_k$ is the thermal state. Now, $\sigma(\nu_k) = \frac{2}{\nu_k+1} \sum_{j=0}^\infty \left(\frac{\nu_k-1}{\nu_k+1}\right)^j |j\rangle\langle j|$. The covariance matrix is $V = S\left[\bigoplus_{k=1}^n \nu_k I_k\right] S^\dagger$[6][eq23, eq32]

Using *Lemma 1* of [6],

$$\rho^p = tr(\rho^p)\rho(p)$$

where $tr(\rho^p) = \prod_{k=1}^n G_p(\nu_k) = \prod_{k=1}^n \frac{2^p}{(\nu_k+1)^p - (\nu_k-1)^p}$. [6][eq30]

Then,

$$Q_s = tr(\hat{\rho}_A^s \hat{\rho}_B^{1-s}) = \prod_{k=1}^n G_s \alpha_k G_{1-s}(\beta_k) tr(\hat{\rho}_A(s)\hat{\rho}_B(1-s)) \tag{99}$$

with covariance matrices $V_A(s)$ and $V_B(1-s)$.

Also, from chapter 2 we have

$$tr(\hat{O}_1, \hat{O}_2) = \int_{\mathbb{C}^n} \frac{d^{2n}\vec{\lambda}}{\pi^n} \chi[\hat{O}_1](\vec{\lambda})\chi[\hat{O}_2](-\vec{\lambda})$$

Using the characteristics function for the gaussian state $\chi(\vec{\lambda}) = exp(-\frac{1}{2}\vec{\lambda}^T\sigma\vec{\lambda} + i\vec{\lambda}^T\bar{R})$ we get,

$$tr(\hat{\rho}_A, \hat{\rho}_B) = \int_{\mathbb{C}^n} \frac{d^{2n}\vec{\lambda}}{\pi^n} exp\left(-\frac{1}{2}\vec{\lambda}^T(V_A + V_B)\vec{\lambda} + i(\bar{x}_A - \bar{x}_B)\vec{\lambda}\right) \tag{100}$$

This is a standard integral. Solving we get,

$$tr(\hat{\rho}_A, \hat{\rho}_B) \propto exp\left(-\frac{1}{2}(\vec{x}_A - \vec{x}_B)^T(V_A + V_B)^{-1}(\vec{x}_A - \vec{x}_B)\right)$$

$$tr(\hat{\rho}_A, \hat{\rho}_B) \propto exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

where $\vec{d} = \vec{x}_A - \vec{x}_B$ Hence,

$$Q_s = tr(\hat{\rho}_A^s\hat{\rho}_B^{1-s}) \propto \prod_{k=1}^{n} G_s\alpha_k G_{1-s}(\beta_k) exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

$$Q_s = tr(\hat{\rho}_A^s\hat{\rho}_B^{1-s}) = \bar{Q}_s exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

It can further be calculated to find the value of $\bar{Q}_s$ as

$$\bar{Q}_s = \frac{2^n \prod_{k=1}^{n} G_s(\alpha_k)G_{1-s}(\beta_k)}{\sqrt{\det\left(V_A(s) + V_B(1-s)\right)}}$$

where,

$$V_A(s) = S_A\left(\bigotimes_{k=1}^{n} \Lambda_s(\alpha_k)\mathbb{I}_k\right)S_A^T$$

$$V_B(1-s) = S_B\left(\bigotimes_{k=1}^{n} \Lambda_{1-s}(\beta_k)\mathbb{I}_k\right)S_B^T$$

$$G_p(x) = \frac{2^p}{(x+1)^p - (x-1)^p}$$

$$\Lambda_p(x) = \frac{(x+1)^p + (x-1)^p}{(x+1)^p - (x-1)^p}$$

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